

CF Q1[ASN-1]

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1. . Three Parts of the Array

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        long[] d = new long[n];
        for (int i = 0; i < n; i++) {
            d[i] = sc.nextLong();
        }
        int left = 0;
        int right = n - 1;
        long leftSum = 0;
        long rightSum = 0;
        long answer = 0;
        while (left <= right) {

            if (leftSum <= rightSum) {
                leftSum += d[left];
                left++;
            } else {
                rightSum += d[right];
                right--;
            }
            if (leftSum == rightSum) {
                answer = leftSum;
            }
        }
        System.out.println(answer);
    }
}
```

2. Boats Competition

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int t = sc.nextInt();

        while (t-- > 0) {
            int n = sc.nextInt();

            int[] freq = new int[n + 1];

            for (int i = 0; i < n; i++) {
                int w = sc.nextInt();
                freq[w]++;
            }

            int answer = 0;

            // Try every possible team weight
            for (int sum = 2; sum <= 2 * n; sum++) {

                int teams = 0;

                for (int weight = 1; weight < sum; weight++) {
                    int other = sum - weight;

                    if (other < 1 || other > n) {
                        continue;
                    }

                    if (weight == other) {
                        teams += freq[weight] / 2;
                    } else {
                        teams += Math.min(freq[weight], freq[other]);
                    }
                }
            }
        }
    }
}
```

```
// Every pair was counted twice
teams /= 2;

answer = Math.max(answer, teams);
}

System.out.println(answer);
}
}
```